cis112
String

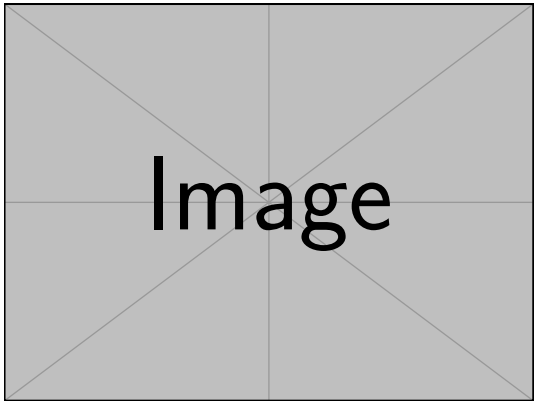
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Content

1. String Object
2. Comparison



String Object

Constructor

Constructor

str-construct.java — Java

```java
System.out.println("\nconstruct");
String s0 = new String();
String s1 = new String("Hello world");
String s2 = "Hello world";
String s3 = "Hello world";
System.out.println("s0:" + s0 + "<");
System.out.println("s1:" + s1 + "<");
System.out.println("s2:" + s2 + "<");
System.out.println("s3:" + s3 + "<");
//
System.out.println("(s0 == s1):" + (s0 == s1))
    ;
System.out.println("(s0 == s2):" + (s0 == s2))
    ;
System.out.println("(s1 == s2):" + (s1 == s2))
    ;
System.out.println("(s2 == s3):" + (s2 == s3))
    ;
```

Output — str-construct.txt

```
construct
s0:<
s1:Hello world<
s2:Hello world<
s3:Hello world<
(s0 == s1):false
(s0 == s2):false
(s1 == s2):false
(s2 == s3):true
```

Methods

Replacing

str-replacing.java — Java

```java
System.out.println("\nreplacing");
String s = "Hello world";
System.out.println("s:\t"//
        + s);
System.out.println("s.replace('l', '@'):\t"//
        + s.replace('l', '@'));
System.out.println("s.replaceFirst(\"l\", \"AA
    \"):\t"//
        + s.replaceFirst("l", "AA"));
System.out.println("s.replaceAll(\"l\", \"AA
    \"):\t"//
        + s.replaceAll("l", "AA"));
```

Output — str-replacing.txt

```
replacing
s:      Hello world
s.replace('l', '@'):    He@@o wor@d
s.replaceFirst("l", "AA"):
    HeAAlo world
s.replaceAll("l", "AA"):
    HeAAAAo worAAd
```

Parsing

Splitting

str-splitting.java — Java

```
1 System.out.println("\nsplitting");
2 String[] token = "Hello world".split("l", 0);
3 System.out.println("token:" + Arrays.toString(
      token));
```

Output — str-splitting.txt

```
splitting
token:[He, , o wor, d]
```

Example: Parsing CSV

str-parsingCsv.java — Java

```java
System.out.println("\nexample_parsingCsv");
String txt = "aa,45,cccc,dd"//
        + "\nee,ffff,gggg"//
        + "\n12,34,789,6543";
//
String[] line = txt.split("[\n]");
for (int i = 0; i < line.length; i++) {
    String[] token = line[i].split("[,]");
    System.out.println("line:" + i + "\t"//
            + Arrays.toString(token));
}
```

Output — str-parsingCsv.txt

```
example_parsingCsv
line:0  [aa, 45, cccc, dd]
line:1  [ee, ffff, gggg]
line:2  [12, 34, 789, 6543]
```

Example: European Number Handler

from EUR7.654.321,1234 to 7654321.1234

str-europeanNumberHandler.java — Java

```
1  System.out.println("\
       nexample_EuropeanNumberHandler");
2  String european = "EUR7.654.321,1234";
3  System.out.println("european:" + european);
4  String tmp = new String(european);
5  tmp = tmp.replaceFirst("EUR", "");
6  System.out.println("replaceFirst(\"EUR\",
       \"\"):" + tmp);
7  tmp = tmp.replaceAll("\\.", "");
8  System.out.println("replaceAll(\"\\\\.\",
       \"\"):" + tmp);
9  tmp = tmp.replaceFirst(",", ".");
10 System.out.println("replaceFirst(\",\", \".\")
       :" + tmp);
11 double d = Double.parseDouble(tmp);
12 System.out.println("(d + 3):" + (d + 3));
13 System.out.println("european:" + european);
```

Output — str-europeanNumberHandler.txt

```
example_EuropeanNumberHandler
european:EUR7.654.321,1234
replaceFirst("EUR", "")
     :7.654.321,1234
replaceAll("\\.", ""):7654321,1234
replaceFirst(",", "."):7654321.1234
(d + 3):7654324.1234
european:EUR7.654.321,1234
```

Regular Expressions

Matching

str-matching.java — Java

```java
System.out.println("\nmatching");
System.out.println("Hello world".matches(".*wor.*"));
System.out.println("Hello abcld".matches(".*wor.*"));
```

Output — str-matching.txt

```
matching
true
false
```

StringBuilder

StringBuilder

str-StringBuilder.java — Java

```java
System.out.println("\nstringBuilder");
StringBuilder s = new StringBuilder();
System.out.println("aaa:" + s + "<");
```

Output — str-StringBuilder.txt

```
stringBuilder
aaa:<
```

Example: toString

str-toString.java — Java

```java
private static void example_toString() {
    System.out.println("\nexample_toString");
    StringsW02 stringsW02 = new StringsW02();
    System.out.println(stringsW02);
}

@Override
public String toString() {
    StringBuilder s = new StringBuilder("[
     StringsW02:");
    s.append("  field1=" + field1);
    s.append(", field2=" + field2);
    s.append("]");
    return s.toString();
}
```

Output — str-toString.txt

```
example_toString
[StringsW02:  field1=valueOfField1,
     field2=valueOfField2]
```

Comparison

Comparison

str-comparison.java — Java

```java
System.out.println("\ncomparison");
String s0="abc";
String s1="aBc";
String s2="abcd";
//
System.out.println("(s0):"+(s0));
System.out.println("(s1):"+(s1));
System.out.println("(s2):"+(s2));
//
System.out.println("(s0.equals(s1)):"+(s0.
    equals(s1)));
System.out.println("(s0.equals(s2)):"+(s0.
    equals(s2)));
System.out.println("(s1.equals(s2)):"+(s1.
    equals(s2)));
//
System.out.println("(s0.equalsIgnoreCase(s1)):
    "+(s0.equalsIgnoreCase(s1)));
System.out.println("(s0.equalsIgnoreCase(s2)):
    "+(s0.equalsIgnoreCase(s2)));
```

Output — str-comparison.txt

```
comparison
(s0):abc
(s1):aBc
(s2):abcd
(s0.equals(s1)):false
(s0.equals(s2)):false
(s1.equals(s2)):false
(s0.equalsIgnoreCase(s1)):true
(s0.equalsIgnoreCase(s2)):false
(s1.equalsIgnoreCase(s2)):false
```

Concatenation

str-concatenation.java — Java

```java
System.out.println("\nconcatenation");
String s = new String("Alice");
System.out.println("s:" + s);
s = s.concat(" and Bob");
System.out.println("s:" + s);
```

Output — str-concatenation.txt

```
concatenation
s:Alice
s:Alice and Bob
```

Some web resources

- The Java Language Specification- Java SE 8 Edition url
  James Gosling, Bill Joy, Guy Steele, Gilad Bracha, Alex Buckley, 2015-02-13.
- The Java Tutorials url
- Strings in The Java Tutorials url

References I